JEE MAIN · MATHEMATICS · CHAPTER-WISE PYQS

Inverse Trigonometric Functions

JEE Main Mathematics · Previous Year Questions with Answer Key

97	91	2002 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

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MCQ (Single Correct Answer)	81 questions
Numerical	15 questions
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Answer key	at the end

1 **Solve**
Print it, or work through it on screen.

2 **Check**
The answer key is on the last pages.

MCQ (Single Correct Answer) 81 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

Let $\alpha = 3\sin^{-1}\left(\frac{6}{11}\right)$ and $\beta = 3\cos^{-1}\left(\frac{4}{9}\right)$, where inverse trigonometric functions take only the principal values.

Given below are two statements :

Statement I : $\cos(\alpha+\beta) > 0$.

Statement II : $\cos(\alpha) < 0$.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Q2. JEE Main 2026 (Online) 6th April Evening Shift

If $\sin\left(\tan^{-1}(x\sqrt{2})\right) = \cot\left(\sin^{-1}\sqrt{1-x^2}\right), x \in (0,1)$, then the value of x is:

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{2}{3}$ (D) $\frac{5}{8}$

Q3. JEE Main 2026 (Online) 6th April Morning Shift

Let $0 < \alpha < 1, \beta = \frac{1}{3\alpha}$ and $\tan^{-1}(1-\alpha) + \tan^{-1}(1-\beta) = \frac{\pi}{4}$. Then $6(\alpha+\beta)$ is equal to:

(A) 6 (B) 7 (C) 8 (D) 9

Q4. JEE Main 2026 (Online) 28th January Evening Shift

Considering the principal values of inverse trigonometric functions, the value of the expression

$\tan\left(2\sin^{-1}\left(\frac{2}{\sqrt{13}}\right) - 2\cos^{-1}\left(\frac{3}{\sqrt{10}}\right)\right)$

is equal to :

(A) $\frac{33}{56}$ (B) $-\frac{33}{56}$ (C) $-\frac{16}{63}$ (D) $\frac{16}{63}$

Q5. JEE Main 2026 (Online) 24th January Evening Shift

If the domain of the function $f(x) = \sin^{-1}\left(\frac{1}{x^2-2x-2}\right)$, is $(-\infty,\alpha] \cup [\beta,\gamma] \cup [\delta,\infty)$, then $\alpha+\beta+\gamma+\delta$ is equal to

(A) 4 (B) 2 (C) 5 (D) 3

Q6. JEE Main 2026 (Online) 22nd January Morning Shift

The number of solutions of $\tan^{-1} 4x + \tan^{-1} 6x = \frac{\pi}{6}$, where $-\frac{1}{2\sqrt{6}} < x < \frac{1}{2\sqrt{6}}$, is equal to :

(A) 2 (B) 0 (C) 3 (D) 1

Q7. JEE Main 2026 (Online) 21st January Morning Shift

If the domain of the function $f(x) = \cos^{-1}\left(\frac{2x-5}{11-3x}\right) + \sin^{-1}\left(2x^2 - 3x + 1\right)$ is the interval $[\alpha, \beta]$, then $\alpha + 2\beta$ is equal to :

(A) 5 (B) 2 (C) 3 (D) 1

Q8. JEE Main 2025 (Online) 8th April Evening Shift

The value of $\cot^{-1}\left(\frac{\sqrt{1+\tan^2(2)}-1}{\tan(2)}\right) - \cot^{-1}\left(\frac{\sqrt{1+\tan^2\left(\frac{1}{2}\right)}+1}{\tan\left(\frac{1}{2}\right)}\right)$ is equal to

(A) $\pi - \frac{3}{2}$ (B) $\pi + \frac{5}{2}$ (C) $\pi - \frac{5}{4}$ (D) $\pi + \frac{3}{2}$

Q9. JEE Main 2025 (Online) 4th April Evening Shift

The sum of the infinite series $\cot^{-1}\left(\frac{7}{4}\right) + \cot^{-1}\left(\frac{19}{4}\right) + \cot^{-1}\left(\frac{39}{4}\right) + \cot^{-1}\left(\frac{67}{4}\right) + \ldots$. is :

(A) $\frac{\pi}{2} + \cot^{-1}\left(\frac{1}{2}\right)$ (B) $\frac{\pi}{2} - \cot^{-1}\left(\frac{1}{2}\right)$

(C) $\frac{\pi}{2} - \tan^{-1}\left(\frac{1}{2}\right)$ (D) $\frac{\pi}{2} + \tan^{-1}\left(\frac{1}{2}\right)$

Q10. JEE Main 2025 (Online) 4th April Morning Shift

Considering the principal values of the inverse trigonometric functions,
$\sin^{-1}\left(\frac{\sqrt{3}}{2}x + \frac{1}{2}\sqrt{1-x^2}\right), -\frac{1}{2} < x < \frac{1}{\sqrt{2}}$, is equal to

(A) $\frac{-5\pi}{6} - \sin^{-1} x$ (B) $\frac{5\pi}{6} - \sin^{-1} x$ (C) $\frac{\pi}{6} + \sin^{-1} x$ (D) $\frac{\pi}{4} + \sin^{-1} x$

Q11. JEE Main 2025 (Online) 28th January Evening Shift

Let [x] denote the greatest integer less than or equal to x. Then the domain of $f(x) = \sec^{-1}(2[x] + 1)$ is:

(A) $(-\infty, \infty)$ (B) $(-\infty, \infty) - \{0\}$ (C) $(-\infty, -1] \cup [0, \infty)$ (D) $(-\infty, -1] \cup [1, \infty)$

Q12. JEE Main 2025 (Online) 28th January Morning Shift

$\cos\left(\sin^{-1}\frac{3}{5} + \sin^{-1}\frac{5}{13} + \sin^{-1}\frac{33}{65}\right)$ is equal to:

(A) $\frac{33}{65}$ (B) 1 (C) $\frac{32}{65}$ (D) 0

Q13. JEE Main 2025 (Online) 24th January Evening Shift

If $\alpha > \beta > \gamma > 0$, then the expression $\cot^{-1}\left\{\beta + \frac{(1+\beta^2)}{(\alpha-\beta)}\right\} + \cot^{-1}\left\{\gamma + \frac{(1+\gamma^2)}{(\beta-\gamma)}\right\} + \cot^{-1}\left\{\alpha + \frac{(1+\alpha^2)}{(\gamma-\alpha)}\right\}$ is equal to :

(A) 3π (B) $\frac{\pi}{2} - (\alpha+\beta+\gamma)$

(C) π (D) 0

Q14. JEE Main 2025 (Online) 23rd January Morning Shift

If $\frac{\pi}{2} \le x \le \frac{3\pi}{4}$, then $\cos^{-1}\left(\frac{12}{13}\cos x + \frac{5}{13}\sin x\right)$ is equal to

(A) $x + \tan^{-1}\frac{5}{12}$ (B) $x - \tan^{-1}\frac{4}{3}$ (C) $x + \tan^{-1}\frac{4}{5}$ (D) $x - \tan^{-1}\frac{5}{12}$

Q15. JEE Main 2025 (Online) 22nd January Morning Shift

Using the principal values of the inverse trigonometric functions, the sum of the maximum and the minimum values of $16\left(\left(\sec^{-1}x\right)^2 + \left(\operatorname{cosec}^{-1}x\right)^2\right)$ is :

(A) $24\pi^2$ (B) $18\pi^2$ (C) $22\pi^2$ (D) $31\pi^2$

Q16. JEE Main 2024 (Online) 4th April Evening Shift

Given that the inverse trigonometric function assumes principal values only. Let x, y be any two real numbers in $[-1, 1]$ such that $\cos^{-1}x - \sin^{-1}y = \alpha$, $\frac{-\pi}{2} \le \alpha \le \pi$. Then, the minimum value of $x^2 + y^2 + 2xy\sin\alpha$ is

(A) 0 (B) -1 (C) $\frac{1}{2}$ (D) $\frac{-1}{2}$

Q17. JEE Main 2024 (Online) 4th April Morning Shift

If the domain of the function $\sin^{-1}\left(\frac{3x-22}{2x-19}\right) + \log_e\left(\frac{3x^2-8x+5}{x^2-3x-10}\right)$ is $(\alpha, \beta]$, then $3\alpha + 10\beta$ is equal to:

(A) 95 (B) 100 (C) 97 (D) 98

Q18. JEE Main 2024 (Online) 31st January Evening Shift

If $a = \sin^{-1}(\sin(5))$ and $b = \cos^{-1}(\cos(5))$, then $a^2 + b^2$ is equal to

(A) 25 (B) $4\pi^2 + 25$ (C) $8\pi^2 - 40\pi + 50$ (D) $4\pi^2 - 20\pi + 50$

Q19. JEE Main 2024 (Online) 31st January Morning Shift

For $\alpha, \beta, \gamma \neq 0$, if $\sin^{-1}\alpha + \sin^{-1}\beta + \sin^{-1}\gamma = \pi$ and $(\alpha+\beta+\gamma)(\alpha-\gamma+\beta) = 3\alpha\beta$, then γ equals

(A) $\sqrt{3}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{\sqrt{3}-1}{2\sqrt{2}}$

Q20. JEE Main 2024 (Online) 29th January Evening Shift

Let $x = \frac{m}{n}$ (m, n are co-prime natural numbers) be a solution of the equation $\cos\left(2\sin^{-1}x\right) = \frac{1}{9}$ and let $\alpha, \beta(\alpha > \beta)$ be the roots of the equation $mx^2 - nx - m + n = 0$. Then the point (α, β) lies on the line

(A) $3x - 2y = -2$ (B) $3x + 2y = 2$ (C) $5x + 8y = 9$ (D) $5x - 8y = -9$

Q21. JEE Main 2024 (Online) 27th January Evening Shift

Considering only the principal values of inverse trigonometric functions, the number of positive real values of x satisfying $\tan^{-1}(x) + \tan^{-1}(2x) = \frac{\pi}{4}$ is :

(A) more than 2 (B) 2 (C) 0 (D) 1

Q22. JEE Main 2023 (Online) 15th April Morning Shift

If the domain of the function

$f(x) = \log_e\left(4x^2 + 11x + 6\right) + \sin^{-1}(4x + 3) + \cos^{-1}\left(\frac{10x+6}{3}\right)$ is $(\alpha, \beta]$, then

$36|\alpha + \beta|$ is equal to :

(A) 72 (B) 54 (C) 45 (D) 63

Q23. JEE Main 2023 (Online) 1st February Evening Shift

Let $S = \left\{x \in R : 0 < x < 1 \text{ and } 2\tan^{-1}\left(\frac{1-x}{1+x}\right) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)\right\}$.

If $\mathrm{n(S)}$ denotes the number of elements in S then :

(A) $\mathrm{n(S)} = 0$

(B) $\mathrm{n(S)} = 2$ and only one element in S is less than $\frac{1}{2}$.

(C) $\mathrm{n(S)} = 1$ and the element in S is more than $\frac{1}{2}$.

(D) $\mathrm{n(S)} = 1$ and the element in S is less than $\frac{1}{2}$.

Q24. JEE Main 2023 (Online) 1st February Morning Shift

Let S be the set of all solutions of the equation $\cos^{-1}(2x) - 2\cos^{-1}\left(\sqrt{1-x^2}\right) = \pi, x \in \left[-\frac{1}{2}, \frac{1}{2}\right]$.

Then $\sum\limits_{x \in S} 2\sin^{-1}\left(x^2 - 1\right)$ is equal to :

(A) $\pi - 2\sin^{-1}\left(\frac{\sqrt{3}}{4}\right)$ (B) $\pi - \sin^{-1}\left(\frac{\sqrt{3}}{4}\right)$

(C) $\frac{-2\pi}{3}$ (D) 0

Q25. JEE Main 2023 (Online) 31st January Evening Shift

Let (a, b) $\subset (0, 2\pi)$ be the largest interval for which $\sin^{-1}(\sin\theta) - \cos^{-1}(\sin\theta) > 0, \theta \in (0, 2\pi)$, holds.

If $\alpha x^2 + \beta x + \sin^{-1}\left(x^2 - 6x + 10\right) + \cos^{-1}\left(x^2 - 6x + 10\right) = 0$ and $\alpha - \beta = b - a$, then α is equal to :

(A) $\frac{\pi}{16}$ (B) $\frac{\pi}{48}$ (C) $\frac{\pi}{8}$ (D) $\frac{\pi}{12}$

Q26. JEE Main 2023 (Online) 31st January Morning Shift

If $\sin^{-1}\frac{\alpha}{17} + \cos^{-1}\frac{4}{5} - \tan^{-1}\frac{77}{36} = 0, 0 < \alpha < 13$, then $\sin^{-1}(\sin\alpha) + \cos^{-1}(\cos\alpha)$ is equal to :

(A) 16 (B) π (C) $16 - 5\pi$ (D) 0

Q27. JEE Main 2023 (Online) 24th January Morning Shift

$\tan^{-1}\left(\frac{1+\sqrt{3}}{3+\sqrt{3}}\right) + \sec^{-1}\left(\sqrt{\frac{8+4\sqrt{3}}{6+3\sqrt{3}}}\right)$ is equal to :

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{4}$

Q28. JEE Main 2022 (Online) 29th July Evening Shift

The domain of the function $f(x) = \sin^{-1}\left(\frac{x^2-3x+2}{x^2+2x+7}\right)$ is :

(A) $[1, \infty)$ (B) $[-1, 2]$ (C) $[-1, \infty)$ (D) $(-\infty, 2]$

Q29. JEE Main 2022 (Online) 28th July Evening Shift

The sum of the absolute maximum and absolute minimum values of the function $f(x) = \tan^{-1}(\sin x - \cos x)$ in the interval $[0, \pi]$ is :

(A) 0 (B) $\tan^{-1}\left(\frac{1}{\sqrt{2}}\right) - \frac{\pi}{4}$

(C) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right) - \frac{\pi}{4}$ (D) $\frac{-\pi}{12}$

Q30. JEE Main 2022 (Online) 28th July Morning Shift

Considering only the principal values of the inverse trigonometric functions, the domain of the function $f(x) = \cos^{-1}\left(\frac{x^2-4x+2}{x^2+3}\right)$ is :

(A) $\left(-\infty, \frac{1}{4}\right]$ (B) $\left[-\frac{1}{4}, \infty\right)$

(C) $(-1/3, \infty)$ (D) $\left(-\infty, \frac{1}{3}\right]$

Q31. JEE Main 2022 (Online) 28th July Morning Shift

Considering the principal values of the inverse trigonometric functions, the sum of all the solutions of the equation $\cos^{-1}(x) - 2\sin^{-1}(x) = \cos^{-1}(2x)$ is equal to :

(A) 0 (B) 1 (C) $\frac{1}{2}$ (D) $-\frac{1}{2}$

Q32. JEE Main 2022 (Online) 27th July Evening Shift

The domain of the function $f(x)=\sin^{-1}\left[2x^2-3\right]+\log_2\left(\log_{\frac{1}{2}}\left(x^2-5x+5\right)\right)$, where [t] is the greatest integer function, is :

(A) $\left(-\sqrt{\frac{5}{2}}, \frac{5-\sqrt{5}}{2}\right)$

(B) $\left(\frac{5-\sqrt{5}}{2}, \frac{5+\sqrt{5}}{2}\right)$

(C) $\left(1, \frac{5-\sqrt{5}}{2}\right)$

(D) $\left[1, \frac{5+\sqrt{5}}{2}\right)$

Q33. JEE Main 2022 (Online) 26th July Evening Shift

If $0<x<\frac{1}{\sqrt{2}}$ and $\frac{\sin^{-1}x}{\alpha}=\frac{\cos^{-1}x}{\beta}$, then the value of $\sin\left(\frac{2\pi\alpha}{\alpha+\beta}\right)$ is :

(A) $4\sqrt{(1-x^2)}\left(1-2x^2\right)$

(B) $4x\sqrt{(1-x^2)}\left(1-2x^2\right)$

(C) $2x\sqrt{(1-x^2)}\left(1-4x^2\right)$

(D) $4\sqrt{(1-x^2)}\left(1-4x^2\right)$

Q34. JEE Main 2022 (Online) 26th July Morning Shift

$\tan\left(2\tan^{-1}\frac{1}{5}+\sec^{-1}\frac{\sqrt{5}}{2}+2\tan^{-1}\frac{1}{8}\right)$ is equal to :

(A) 1

(B) 2

(C) $\frac{1}{4}$

(D) $\frac{5}{4}$

Q35. JEE Main 2022 (Online) 30th June Morning Shift

Let m and M respectively be the minimum and the maximum values of $f(x)=\sin^{-1}2x+\sin 2x+\cos^{-1}2x+\cos 2x,\ x\in\left[0,\frac{\pi}{8}\right]$. Then m + M is equal to :

(A) $1+\sqrt{2}+\pi$

(B) $\left(1+\sqrt{2}\right)\pi$

(C) $\pi+\sqrt{2}$

(D) $1+\pi$

Q36. JEE Main 2022 (Online) 30th June Morning Shift

Let $\alpha=\tan\left(\frac{5\pi}{16}\sin\left(2\cos^{-1}\left(\frac{1}{\sqrt{5}}\right)\right)\right)$ and $\beta=\cos\left(\sin^{-1}\left(\frac{4}{5}\right)+\sec^{-1}\left(\frac{5}{3}\right)\right)$ where the inverse trigonometric functions take principal values. Then, the equation whose roots are α and β is :

(A) $15x^2-8x-7=0$

(B) $5x^2-12x+7=0$

(C) $25x^2-18x-7=0$

(D) $25x^2-32x+7=0$

Q37. JEE Main 2022 (Online) 29th June Morning Shift

The domain of the function $\cos^{-1}\left(\frac{2\sin^{-1}\left(\frac{1}{4x^2-1}\right)}{\pi}\right)$ is :

(A) $R-\left\{-\frac{1}{2},\frac{1}{2}\right\}$

(B) $(-\infty,-1]\cup[1,\infty)\cup\{0\}$

(C) $\left(-\infty,\frac{-1}{2}\right)\cup\left(\frac{1}{2},\infty\right)\cup\{0\}$

(D) $\left(-\infty,\frac{-1}{\sqrt{2}}\right]\cup\left[\frac{1}{\sqrt{2}},\infty\right)\cup\{0\}$

Q38. JEE Main 2022 (Online) 27th June Evening Shift

The value of $\cot\left(\sum_{n=1}^{50}\tan^{-1}\left(\frac{1}{1+n+n^2}\right)\right)$ is :

(A) $\frac{26}{25}$ (B) $\frac{25}{26}$ (C) $\frac{50}{51}$ (D) $\frac{52}{51}$

Q39. JEE Main 2022 (Online) 27th June Morning Shift

$\sin^{-1}\left(\sin\frac{2\pi}{3}\right)+\cos^{-1}\left(\cos\frac{7\pi}{6}\right)+\tan^{-1}\left(\tan\frac{3\pi}{4}\right)$ is equal to :

(A) $\frac{11\pi}{12}$ (B) $\frac{17\pi}{12}$

(C) $\frac{31\pi}{12}$ (D) $-\frac{3\pi}{4}$

Q40. JEE Main 2022 (Online) 26th June Evening Shift

If the inverse trigonometric functions take principal values then

$\cos^{-1}\left(\frac{3}{10}\cos\left(\tan^{-1}\left(\frac{4}{3}\right)\right)+\frac{2}{5}\sin\left(\tan^{-1}\left(\frac{4}{3}\right)\right)\right)$ is equal to :

(A) 0 (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{6}$

Q41. JEE Main 2022 (Online) 25th June Evening Shift

The value of $\tan^{-1}\left(\frac{\cos\left(\frac{15\pi}{4}\right)-1}{\sin\left(\frac{\pi}{4}\right)}\right)$ is equal to :

(A) $-\frac{\pi}{4}$ (B) $-\frac{\pi}{8}$ (C) $-\frac{5\pi}{12}$ (D) $-\frac{4\pi}{9}$

Q42. JEE Main 2022 (Online) 24th June Evening Shift

Let $x*y=x^2+y^3$ and $(x*1)*1=x*(1*1)$.

Then a value of $2\sin^{-1}\left(\frac{x^4+x^2-2}{x^4+x^2+2}\right)$ is :

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{6}$

Q43. JEE Main 2022 (Online) 24th June Morning Shift

The set of all values of k for which

$(\tan^{-1}x)^3+(\cot^{-1}x)^3=k\pi^3,\ x\in R$, is the interval :

(A) $\left[\frac{1}{32},\frac{7}{8}\right)$ (B) $\left(\frac{1}{24},\frac{13}{16}\right)$

(C) $\left[\frac{1}{48},\frac{13}{16}\right]$ (D) $\left[\frac{1}{32},\frac{9}{8}\right)$

Q44. JEE Main 2022 (Online) 24th June Morning Shift

The domain of the function

$f(x)=\frac{\cos^{-1}\left(\frac{x^2-5x+6}{x^2-9}\right)}{\log_e(x^2-3x+2)}$ is :

(A) $(-\infty,1)\cup(2,\infty)$

(B) $(2,\infty)$

(C) $\left[-\frac{1}{2},1\right)\cup(2,\infty)$

(D) $\left[-\frac{1}{2},1\right)\cup(2,\infty)-\left\{3,\frac{3+\sqrt{5}}{2},\frac{3-\sqrt{5}}{2}\right\}$

Q45. JEE Main 2021 (Online) 1st September Evening Shift

$\cos^{-1}(\cos(-5))+\sin^{-1}(\sin(6))-\tan^{-1}(\tan(12))$ is equal to :

(The inverse trigonometric functions take the principal values)

(A) $3\pi-11$ (B) $4\pi-9$ (C) $4\pi-11$ (D) $3\pi+1$

Q46. JEE Main 2021 (Online) 31st August Evening Shift

The domain of the function

$f(x)=\sin^{-1}\left(\frac{3x^2+x-1}{(x-1)^2}\right)+\cos^{-1}\left(\frac{x-1}{x+1}\right)$ is :

(A) $\left[0,\frac{1}{4}\right]$ (B) $[-2,0]\cup\left[\frac{1}{4},\frac{1}{2}\right]$

(C) $\left[\frac{1}{4},\frac{1}{2}\right]\cup\{0\}$ (D) $\left[0,\frac{1}{2}\right]$

Q47. JEE Main 2021 (Online) 27th August Evening Shift

Let M and m respectively be the maximum and minimum values of the function f(x) = $\tan^{-1}$ (sin x + cos x) in $\left[0,\frac{\pi}{2}\right]$, then the value of tan(M − m) is equal to :

(A) $2+\sqrt{3}$ (B) $2-\sqrt{3}$ (C) $3+2\sqrt{2}$ (D) $3-2\sqrt{2}$

Q48. JEE Main 2021 (Online) 27th August Morning Shift

If $(\sin^{-1}x)^2-(\cos^{-1}x)^2=a$; 0 < x < 1, a $\neq$ 0, then the value of $2x^2-1$ is :

(A) $\cos\left(\frac{4a}{\pi}\right)$ (B) $\sin\left(\frac{2a}{\pi}\right)$

(C) $\cos\left(\frac{2a}{\pi}\right)$ (D) $\sin\left(\frac{4a}{\pi}\right)$

Q49. JEE Main 2021 (Online) 26th August Evening Shift

The domain of the function $\operatorname{cosec}^{-1}\left(\frac{1+x}{x}\right)$ is :

(A) $\left(-1,-\frac{1}{2}\right]\cup(0,\infty)$ (B) $\left[-\frac{1}{2},0\right)\cup[1,\infty)$

(C) $\left(-\frac{1}{2},\infty\right)-\{0\}$ (D) $\left[-\frac{1}{2},\infty\right)-\{0\}$

Q50. JEE Main 2021 (Online) 26th August Evening Shift

If $\sum_{r=1}^{50} \tan^{-1}\frac{1}{2r^2} = p$, then the value of tan p is :

(A) $\frac{101}{102}$ (B) $\frac{50}{51}$ (C) 100 (D) $\frac{51}{50}$

Q51. JEE Main 2021 (Online) 22th July Evening Shift

If the domain of the function $f(x) = \frac{\cos^{-1}\sqrt{x^2-x+1}}{\sqrt{\sin^{-1}\left(\frac{2x-1}{2}\right)}}$ is the interval $(\alpha, \beta]$, then $\alpha + \beta$ is equal to :

(A) $\frac{3}{2}$ (B) 2 (C) $\frac{1}{2}$ (D) 1

Q52. JEE Main 2021 (Online) 20th July Evening Shift

The value of $\tan\left(2\tan^{-1}\left(\frac{3}{5}\right) + \sin^{-1}\left(\frac{5}{13}\right)\right)$ is equal to :

(A) $\frac{-181}{69}$ (B) $\frac{220}{21}$ (C) $\frac{-291}{76}$ (D) $\frac{151}{63}$

Q53. JEE Main 2021 (Online) 20th July Morning Shift

The number of real roots of the equation $\tan^{-1}\sqrt{x(x+1)} + \sin^{-1}\sqrt{x^2+x+1} = \frac{\pi}{4}$ is :

(A) 1 (B) 2 (C) 4 (D) 0

Q54. JEE Main 2021 (Online) 17th March Evening Shift

The number of solutions of the equation

$\sin^{-1}\left[x^2 + \frac{1}{3}\right] + \cos^{-1}\left[x^2 - \frac{2}{3}\right] = x^2$, for x$\in$[$-$1, 1], and [x] denotes the greatest integer less than or equal to x, is :

(A) 0 (B) Infinite (C) 2 (D) 4

Q55. JEE Main 2021 (Online) 17th March Morning Shift

The sum of possible values of x for

$\tan^{-1}(x+1) + \cot^{-1}\left(\frac{1}{x-1}\right) = \tan^{-1}\left(\frac{8}{31}\right)$ is :

(A) $-\frac{32}{4}$ (B) $-\frac{33}{4}$

(C) $-\frac{31}{4}$ (D) $-\frac{30}{4}$

Q56. JEE Main 2021 (Online) 17th March Morning Shift

If $\cot^{-1}(\alpha) = \cot^{-1} 2 + \cot^{-1} 8 + \cot^{-1} 18 + \cot^{-1} 32 +$ ..... upto 100 terms, then α is :

(A) 1.02 (B) 1.03 (C) 1.01 (D) 1.00

Q57. JEE Main 2021 (Online) 16th March Evening Shift

Given that the inverse trigonometric functions take principal values only. Then, the number of real values of x which satisfy

$\sin^{-1}\left(\frac{3x}{5}\right)+\sin^{-1}\left(\frac{4x}{5}\right)=\sin^{-1}x$ is equal to :

(A) 2 (B) 0 (C) 3 (D) 1

Q58. JEE Main 2021 (Online) 26th February Evening Shift

If 0 < a, b < 1, and $\tan^{-1}a + \tan^{-1}b = \frac{\pi}{4}$, then the value of

$(a+b)-\left(\frac{a^2+b^2}{2}\right)+\left(\frac{a^3+b^3}{3}\right)-\left(\frac{a^4+b^4}{4}\right)+\ldots\ldots$ is :

(A) $\log_e 2$ (B) e

(C) $\log_e\left(\frac{e}{2}\right)$ (D) e^2-1

Q59. JEE Main 2021 (Online) 26th February Morning Shift

If $\frac{\sin^{-1}x}{a}=\frac{\cos^{-1}x}{b}=\frac{\tan^{-1}y}{c}; 0<x<1,$
then the value of $\cos\left(\frac{\pi c}{a+b}\right)$ is :

(A) $\frac{1-y^2}{2y}$ (B) $\frac{1-y^2}{y\sqrt{y}}$ (C) $1-y^2$ (D) $\frac{1-y^2}{1+y^2}$

Q60. JEE Main 2021 (Online) 25th February Evening Shift

$\text{cosec}\left[2\cot^{-1}(5)+\cos^{-1}\left(\frac{4}{5}\right)\right]$ is equal to :

(A) $\frac{75}{56}$ (B) $\frac{65}{56}$ (C) $\frac{56}{33}$ (D) $\frac{65}{33}$

Q61. JEE Main 2021 (Online) 24th February Evening Shift

A possible value of $\tan\left(\frac{1}{4}\sin^{-1}\frac{\sqrt{63}}{8}\right)$ is :

(A) $\sqrt{7}-1$ (B) $\frac{1}{\sqrt{7}}$ (C) $2\sqrt{2}-1$ (D) $\frac{1}{2\sqrt{2}}$

Q62. JEE Main 2020 (Online) 5th September Morning Slot

If S is the sum of the first 10 terms of the series

$\tan^{-1}\left(\frac{1}{3}\right)+\tan^{-1}\left(\frac{1}{7}\right)+\tan^{-1}\left(\frac{1}{13}\right)+\tan^{-1}\left(\frac{1}{21}\right)+\ldots.$

then tan(S) is equal to :

(A) $\frac{10}{11}$ (B) $\frac{5}{11}$ (C) $-\frac{6}{5}$ (D) $\frac{5}{6}$

Q63. JEE Main 2020 (Online) 3rd September Morning Slot

2π - $\left(\sin^{-1}\frac{4}{5}+\sin^{-1}\frac{5}{13}+\sin^{-1}\frac{16}{65}\right)$ is equal to :

(A) $\frac{7\pi}{4}$ (B) $\frac{5\pi}{4}$ (C) $\frac{3\pi}{2}$ (D) $\frac{\pi}{2}$

Q64. JEE Main 2020 (Online) 2nd September Morning Slot

The domain of the function
$f(x)=\sin^{-1}\left(\frac{|x|+5}{x^2+1}\right)$ is $(-\infty,-a]\cup[a,\infty)$. Then a is equal to :

(A) $\frac{\sqrt{17}-1}{2}$ (B) $\frac{1+\sqrt{17}}{2}$ (C) $\frac{\sqrt{17}}{2}+1$ (D) $\frac{\sqrt{17}}{2}$

Q65. JEE Main 2019 (Online) 12th April Morning Slot

The value of $\sin^{-1}\left(\frac{12}{13}\right)-\sin^{-1}\left(\frac{3}{5}\right)$ is equal to :

(A) $\pi-\sin^{-1}\left(\frac{63}{65}\right)$ (B) $\frac{\pi}{2}-\sin^{-1}\left(\frac{56}{65}\right)$
(C) $\frac{\pi}{2}-\cos^{-1}\left(\frac{9}{65}\right)$ (D) $\pi-\cos^{-1}\left(\frac{33}{65}\right)$

Q66. JEE Main 2019 (Online) 10th April Evening Slot

If $\cos^{-1}x-\cos^{-1}\frac{y}{2}=\alpha$,where $-1\le x\le 1$, $-2\le y\le 2$, $x\le\frac{y}{2}$, then for all x, y, $4x^2-4xy\cos\alpha+y^2$ is equal to :

(A) $4\sin^2\alpha$ (B) $2\sin^2\alpha$
(C) $4\sin^2\alpha-2x^2y^2$ (D) $4\cos^2\alpha+2x^2y^2$

Q67. JEE Main 2019 (Online) 8th April Morning Slot

If $\alpha=\cos^{-1}\left(\frac{3}{5}\right)$, $\beta=\tan^{-1}\left(\frac{1}{3}\right)$ where $0<\alpha,\beta<\frac{\pi}{2}$, then $\alpha-\beta$ is equal to :

(A) $\tan^{-1}\left(\frac{9}{14}\right)$ (B) $\sin^{-1}\left(\frac{9}{5\sqrt{10}}\right)$
(C) $\cos^{-1}\left(\frac{9}{5\sqrt{10}}\right)$ (D) $\tan^{-1}\left(\frac{9}{5\sqrt{10}}\right)$

Q68. JEE Main 2019 (Online) 12th January Morning Slot

Considering only the principal values of inverse functions, the set
$A=\{x\ge 0: \tan^{-1}(2x)+\tan^{-1}(3x)=\frac{\pi}{4}\}$

(A) contains two elements (B) contains more than two elements
(C) is an empty set (D) is a singleton

Q69. JEE Main 2019 (Online) 11th January Evening Slot

All x satisfying the inequality $(\cot^{-1}x)^2-7(\cot^{-1}x)+10>0$, lie in the interval :

(A) $(\cot 2,\infty)$ (B) $(-\infty,\cot 5)\cup(\cot 2,\infty)$
(C) $(\cot 5,\cot 4)$ (D) $(-\infty,\cot 5)\cup(\cot 4,\cot 2)$

Q70. JEE Main 2019 (Online) 10th January Evening Slot

The value of $\cot\left(\sum_{n=1}^{19}\cot^{-1}\left(1+\sum_{p=1}^{n}2p\right)\right)$ is :

(A) $\frac{22}{23}$ (B) $\frac{23}{22}$ (C) $\frac{21}{19}$ (D) $\frac{19}{21}$

Q71. JEE Main 2019 (Online) 9th January Evening Slot

If x = $\sin^{-1}(\sin 10)$ and y = $\cos^{-1}(\cos 10)$, then y $-$ x is equal to :

(A) 0 (B) 10 (C) 7π (D) π

Q72. JEE Main 2019 (Online) 9th January Morning Slot

If $\cos^{-1}\left(\frac{2}{3x}\right)+\cos^{-1}\left(\frac{3}{4x}\right)=\frac{\pi}{2}$ (x > $\frac{3}{4}$), then x is equal to :

(A) $\frac{\sqrt{145}}{10}$ (B) $\frac{\sqrt{145}}{11}$ (C) $\frac{\sqrt{145}}{12}$ (D) $\frac{\sqrt{146}}{12}$

Q73. JEE Main 2017 (Online) 9th April Morning Slot

A value of x satisfying the equation sin[$\cot^{-1}$ (1+ x)] = cos [$\tan^{-1}$ x], is :

(A) $-\frac{1}{2}$ (B) -1 (C) 0 (D) $\frac{1}{2}$

Q74. JEE Main 2017 (Online) 8th April Morning Slot

The value of $\tan^{-1}\left[\frac{\sqrt{1+x^2}+\sqrt{1-x^2}}{\sqrt{1+x^2}-\sqrt{1-x^2}}\right]$, $|x|<\frac{1}{2}, x\neq 0$, is equal to :

(A) $\frac{\pi}{4}+\frac{1}{2}\cos^{-1}x^2$ (B) $\frac{\pi}{4}+\cos^{-1}x^2$

(C) $\frac{\pi}{4}-\frac{1}{2}\cos^{-1}x^2$ (D) $\frac{\pi}{4}-\cos^{-1}x^2$

Q75. JEE Main 2015 (Offline)

Let $\tan^{-1}y=\tan^{-1}x+\tan^{-1}\left(\frac{2x}{1-x^2}\right)$,
where $|x|<\frac{1}{\sqrt{3}}$. Then a value of y is :

(A) $\frac{3x-x^3}{1+3x^2}$ (B) $\frac{3x+x^3}{1+3x^2}$

(C) $\frac{3x-x^3}{1-3x^2}$ (D) $\frac{3x+x^3}{1-3x^2}$

Q76. JEE Main 2013 (Offline)

If x, y, z are in A.P. and $\tan^{-1}x, \tan^{-1}y$ and $\tan^{-1}z$ are also in A.P., then :

(A) $x=y=z$ (B) $2x=3y=6z$ (C) $6x=3y=2z$ (D) $6x=4y=3z$

Q77. AIEEE 2008

The value of $\cot\left(\cos ec^{-1}\frac{5}{3}+\tan^{-1}\frac{2}{3}\right)$ is :

(A) $\frac{6}{17}$ (B) $\frac{3}{17}$ (C) $\frac{4}{17}$ (D) $\frac{5}{17}$

Q78. AIEEE 2007

If $\sin^{-1}\left(\frac{x}{5}\right)+\text{cosec}^{-1}\left(\frac{5}{4}\right)=\frac{\pi}{2}$, then the value of x is :

(A) 4 (B) 5 (C) 1 (D) 3

Q79. AIEEE 2005

If $\cos^{-1}x-\cos^{-1}\frac{y}{2}=\alpha$, then $4x^2-4xy\cos\alpha+y^2$ is equal to :

(A) $2\sin 2\alpha$ (B) 4 (C) $4\sin^2\alpha$ (D) $-4\sin^2\alpha$

Q80. AIEEE 2003

The trigonometric equation $\sin^{-1}x=2\sin^{-1}a$ has a solution for :

(A) $|a|\geq\frac{1}{\sqrt{2}}$ (B) $\frac{1}{2}<|a|<\frac{1}{\sqrt{2}}$

(C) all real values of a (D) $|a|\leq\frac{1}{\sqrt{2}}$

Q81. AIEEE 2002

$\cot^{-1}\left(\sqrt{\cos\alpha}\right)-\tan^{-1}\left(\sqrt{\cos\alpha}\right)=x$, then sin x is equal to :

(A) $\tan^2\left(\frac{\alpha}{2}\right)$ (B) $\cot^2\left(\frac{\alpha}{2}\right)$

(C) $\tan\alpha$ (D) $\cot\left(\frac{\alpha}{2}\right)$

Numerical 15 questions

Q82. JEE Main 2026 (Online) 5th April Morning Shift

If $\frac{\pi}{4}+\sum_{p=1}^{11}\tan^{-1}\left(\frac{2^{p-1}}{1+2^{2p-1}}\right)=\alpha$, then $\tan\alpha$ is equal to ____.

Q83. JEE Main 2026 (Online) 28th January Morning Shift

If $k=\tan\left(\frac{\pi}{4}+\frac{1}{2}\cos^{-1}\left(\frac{2}{3}\right)\right)+\tan\left(\frac{1}{2}\sin^{-1}\left(\frac{2}{3}\right)\right)$, then

the number of solutions of the equation $\sin^{-1}(kx-1)=\sin^{-1}x-\cos^{-1}x$ is ____.

Q84. JEE Main 2026 (Online) 21st January Evening Shift

Let the maximum value of $\left(\sin^{-1} x\right)^2 + \left(\cos^{-1} x\right)^2$ for $x \in \left[-\frac{\sqrt{3}}{2}, \frac{1}{\sqrt{2}}\right]$ be $\frac{\mathrm{m}}{\mathrm{n}}\pi^2$, where $\gcd(\mathrm{m}, \mathrm{n}) = 1$. Then $\mathrm{m} + \mathrm{n}$ is equal to ____。

Q85. JEE Main 2025 (Online) 2nd April Evening Shift

If $y = \cos\left(\frac{\pi}{3} + \cos^{-1}\frac{x}{2}\right)$, then $(x-y)^2 + 3y^2$ is equal to _____.

Q86. JEE Main 2025 (Online) 29th January Morning Shift

Let S = $\left\{x : \cos^{-1} x = \pi + \sin^{-1} x + \sin^{-1}(2x+1)\right\}$. Then $\sum\limits_{x \in S}(2x-1)^2$ is equal to _____.

Q87. JEE Main 2025 (Online) 24th January Morning Shift

If for some $\alpha, \beta; \alpha \leq \beta, \alpha + \beta = 8$ and $\sec^2\left(\tan^{-1}\alpha\right) + \operatorname{cosec}^2\left(\cot^{-1}\beta\right) = 36$, then $\alpha^2 + \beta$ is _____

Q88. JEE Main 2024 (Online) 9th April Evening Shift

Let the inverse trigonometric functions take principal values. The number of real solutions of the equation $2\sin^{-1} x + 3\cos^{-1} x = \frac{2\pi}{5}$, is _____.

Q89. JEE Main 2024 (Online) 6th April Morning Shift

For $n \in \mathrm{N}$, if $\cot^{-1} 3 + \cot^{-1} 4 + \cot^{-1} 5 + \cot^{-1} n = \frac{\pi}{4}$, then n is equal to _____.

Q90. JEE Main 2023 (Online) 13th April Evening Shift

For $x \in (-1, 1]$, the number of solutions of the equation $\sin^{-1} x = 2\tan^{-1} x$ is equal to _____.

Q91. JEE Main 2023 (Online) 13th April Morning Shift

If $S = \left\{x \in \mathbb{R} : \sin^{-1}\left(\frac{x+1}{\sqrt{x^2+2x+2}}\right) - \sin^{-1}\left(\frac{x}{\sqrt{x^2+1}}\right) = \frac{\pi}{4}\right\}$, then
$\sum\limits_{x \in s}\left(\sin\left(\left(x^2+x+5\right)\frac{\pi}{2}\right) - \cos\left(\left(x^2+x+5\right)\pi\right)\right)$ is equal to _____.

Q92. JEE Main 2023 (Online) 10th April Evening Shift

If the domain of the function $f(x)=\sec^{-1}\left(\frac{2x}{5x+3}\right)$ is $[\alpha,\beta)\mathrm{U}(\gamma,\delta]$, then $|3\alpha+10(\beta+\gamma)+21\delta|$ is equal to _____.

Q93. JEE Main 2023 (Online) 25th January Morning Shift

If the sum of all the solutions of $\tan^{-1}\left(\frac{2x}{1-x^2}\right)+\cot^{-1}\left(\frac{1-x^2}{2x}\right)=\frac{\pi}{3}, -1<x<1, x\neq 0$, is $\alpha-\frac{4}{\sqrt{3}}$, then α is equal to _____.

Q94. JEE Main 2022 (Online) 27th July Morning Shift

For $k\in\mathbb{R}$, let the solutions of the equation $\cos\left(\sin^{-1}\left(x\cot\left(\tan^{-1}\left(\cos\left(\sin^{-1}x\right)\right)\right)\right)\right)=k, 0<|x|<\frac{1}{\sqrt{2}}$ be α and β, where the inverse trigonometric functions take only principal values. If the solutions of the equation $x^2-bx-5=0$ are $\frac{1}{\alpha^2}+\frac{1}{\beta^2}$ and $\frac{\alpha}{\beta}$, then $\frac{b}{k^2}$ is equal to _____.

Q95. JEE Main 2022 (Online) 25th July Evening Shift

Let $x=\sin(2\tan^{-1}\alpha)$ and $y=\sin\left(\frac{1}{2}\tan^{-1}\frac{4}{3}\right)$. If $S=\{\alpha\in R: y^2=1-x\}$, then $\sum\limits_{\alpha\in S}16\alpha^3$ is equal to _____.

Q96. JEE Main 2022 (Online) 29th June Morning Shift

$50\tan\left(3\tan^{-1}\left(\frac{1}{2}\right)+2\cos^{-1}\left(\frac{1}{\sqrt{5}}\right)\right)+4\sqrt{2}\tan\left(\frac{1}{2}\tan^{-1}(2\sqrt{2})\right)$ is equal to _____.

MCQ (More than One Correct Answer) 1 questions

Q97. JEE Main 2023 (Online) 30th January Evening Shift

Let $a_1=1, a_2, a_3, a_4, \ldots..$ be consecutive natural numbers.

Then $\tan^{-1}\left(\frac{1}{1+a_1a_2}\right)+\tan^{-1}\left(\frac{1}{1+a_2a_3}\right)+\ldots..+\tan^{-1}\left(\frac{1}{1+a_{2021}a_{2022}}\right)$ is equal to :

(A) $\frac{\pi}{4}-\cot^{-1}(2022)$ (B) $\frac{\pi}{4}-\tan^{-1}(2022)$ (C) $\cot^{-1}(2022)-\frac{\pi}{4}$ (D) $\tan^{-1}(2022)-\frac{\pi}{4}$

Answer Key

MCQ (Single Correct Answer)

1. **A**	2. **A**	3. **B**	4. **A**	5. **A**
6. **D**	7. **C**	8. **C**	9. **C**	10. **C**
11. **A**	12. **D**	13. **C**	14. **D**	15. **C**
16. **A**	17. **C**	18. **C**	19. **B**	20. **C**
21. **D**	22. **C**	23. **D**	24. **D**	25. **D**
26. **B**	27. **B**	28. **C**	29. **C**	30. **B**
31. **A**	32. **C**	33. **B**	34. **B**	35. **A**
36. **C**	37. **D**	38. **A**	39. **A**	40. **C**
41. **B**	42. **B**	43. **A**	44. **D**	45. **C**
46. **C**	47. **D**	48. **B**	49. **D**	50. **B**
51. **A**	52. **B**	53. **D**	54. **A**	55. **A**
56. **C**	57. **C**	58. **A**	59. **D**	60. **B**
61. **B**	62. **D**	63. **C**	64. **B**	65. **B**
66. **A**	67. **B**	68. **D**	69. **A**	70. **C**
71. **D**	72. **C**	73. **A**	74. **A**	75. **C**
76. **A**	77. **A**	78. **D**	79. **C**	80. **D**
81. **A**				

Numerical

82. **2048**	83. **1**	84. **65**	85. **3**	86. **5**
87. **14**	88. **0**	89. **47**	90. **2**	91. **4**
92. **24**	93. **2**	94. **12**	95. **130**	96. **29**

MCQ (More than One Correct Answer)

97. **A, D**